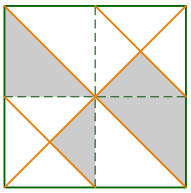
# MCC Entrance Test, August 2026 — Day 2

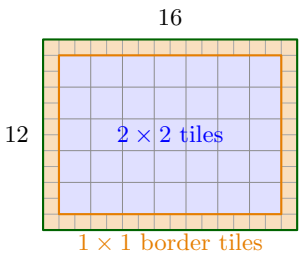
Level T1

**Test Information. Level T1** — 90 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Number Theory (×2), Geometry (×2), Algebra, Combinatorics.
Rules: show your work (partial credit rewards genuine progress); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

**Problem 1 (10 pts).** A counting number is *marked* if it has an *odd* number of factors. (*For example, the number* 6 *has exactly four factors* – 1, 2, 3, *and* 6 – *so* 6 *is not marked.*) How many counting numbers less than 20 are marked?

**Problem 2 (10 pts).** A large square is divided into four smaller squares of equal size by its two midlines, and both full diagonals of the large square are drawn. In two of the small squares (diagonally opposite each other), only one diagonal passes through, splitting each into 2 triangles. In the other two small squares, a second diagonal is also drawn, splitting each of *those* into 4 triangles. In each small square, exactly one triangle is shaded, as shown. What fraction of the large square's area is shaded?






**Problem 3 (15 pts).** Tyler is tiling the floor of a 12-foot by 16-foot rectangular room. He plans to place 1-foot by 1-foot square tiles to form a border, one tile wide, running along all four edges of the room. He will then fill in the rest of the floor using 2-foot by 2-foot square tiles.

(a) What are the dimensions of the inner rectangle that will be covered with the 2-by-2 tiles? [7 pts]

(b) How many tiles in total (both the 1-by-1 border tiles and the 2-by-2 interior tiles together) will Tyler use to cover the whole floor? [8 pts]

**Problem 4 (15 pts).** A vendor sells three kinds of tickets: Type A tickets for \$3 each, Type B tickets for \$1.50 each, and Type C tickets for \$0.50 each. One evening, the vendor sells exactly 20 tickets in total – at least one of each kind – and collects exactly \$20 in all.

(a) Show that if $b$ is the number of Type A tickets sold and $m$ is the number of Type B tickets sold, then $5b + 2m = 20$. [5 pts]

(b) Find how many Type A tickets the vendor sold, and how many dollars this brought in. [10 pts]

**Problem 5 (25 pts).** A survey asks each of 425 people which of three activities – A, B, and C – they have tried. In total, 351 people tried activity A, 71 tried activity B, and 203 tried activity C. Of these people, 199 tried more than one activity, and 8 tried none of the three activities at all.

(a) How many of the 425 people tried at least one of the three activities? [5 pts]

(b) If $a$ people tried exactly two activities and $t$ people tried all three activities, write the equation relating $a$, $t$, and the 199 figure above. [3 pts]

(c) Find $t$, the number of people who tried all three activities. [17 pts]

**Problem 6 (25 pts).** Six contestants each compete in two rounds. Their round-1 scores are $23, 24, 26, 28, 28, 29$ – one score per contestant. Their round-2 scores are $10, 10, 12, 12, 14, 16$ – again one per contestant, but we are not told which round-2 score belongs to which contestant. The winner is whichever contestant has the highest combined score (round 1 + round 2).

(a) Add up all twelve scores to find the total of every contestant's combined score, added together. Use this total to explain why at least one contestant must have a combined score of 39 or more. [7 pts]

(b) Among the round-2 scores $(10, 10, 12, 12, 14, 16)$, how many are 11 or less? [2 pts]

(c) Show that a winning score of exactly 39 is impossible, and find the least possible combined score the winner could have – along with one matching of round-1 and round-2 scores that achieves it. [16 pts]